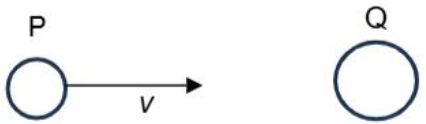
9 A moving object P approaches head-on a stationary object Q with velocity v, as shown.



The two objects collide elastically.

After collision Q moves to the right with speed 0.4 v.

What is the speed of P after the collision?

	A	0	**B**	0.4 v	**C**	0.6 v	**D**	v

L1 **Answer: C**

Let the speed of P after the collision be V.

Take right as positive direction.

For **elastic** collision,
relative speed of approach = relative speed of separation
$v - 0 = 0.4\,v - V$
$V = -0.6\,v$
∴Speed of P = 0.6 v